

Executive Business Report

Sales Forecasting and Demand Analysis for Retail Superstore

Executive Summary

This project analyzed four years of retail sales data to understand sales patterns, forecast future demand, detect unusual sales behavior, and segment products based on demand characteristics. Three forecasting models—SARIMA, Prophet, and XGBoost—were developed and compared. Based on forecasting accuracy, XGBoost performed the best and was selected as the recommended forecasting model. The results provide valuable insights that can help improve inventory planning, reduce stock shortages, minimize excess inventory, and support data-driven business decisions.

Key Findings

The exploratory analysis revealed several important business insights. Technology products generated the highest overall revenue, while Office Supplies maintained relatively stable sales throughout the observed period. Furniture sales showed greater fluctuations, indicating more variable customer demand.

Among all regions, the West region recorded the strongest sales performance, whereas the East region demonstrated consistent growth over the years. Analysis of shipping performance indicated that delivery times remained fairly consistent across regions, suggesting an efficient logistics network.

The sales trend also showed clear seasonality. Sales increased significantly during November and December, likely due to holiday shopping and promotional campaigns, while the beginning of each year experienced comparatively lower demand.

Forecasting Results

Three forecasting approaches were evaluated using monthly sales data. After comparing their prediction accuracy, the XGBoost model achieved the best overall performance and was selected as the final forecasting model.

The three-month forecast suggests that sales are expected to remain relatively stable over the coming months with normal seasonal variations. Based on the model's historical

accuracy, the predicted sales values provide a reliable estimate for inventory and procurement planning. However, actual sales may vary due to unexpected events such as promotional campaigns, market changes, or supply chain disruptions.

Anomaly Detection

Two anomaly detection techniques, Isolation Forest and Rolling Z-Score, were used to identify unusually high or low sales weeks. Both methods detected several common anomalies, increasing confidence in the results.

Most unusually high sales periods are likely associated with seasonal promotions, festive shopping, and year-end discount campaigns. On the other hand, unusually low sales weeks may be explained by seasonal slowdowns, temporary inventory shortages, or reduced customer demand. These findings can help management investigate unusual business events and improve future planning.

Product Demand Segmentation

Products were grouped into demand-based segments using K-Means clustering. The analysis identified products with stable demand, products experiencing growth, highly volatile products, and premium products with lower purchasing frequency.

Based on these segments, stable products should maintain consistent inventory levels, growing products should receive gradually increased stock, volatile products should be monitored carefully with limited safety stock, and premium products should follow a demand-driven replenishment strategy to reduce inventory holding costs.

Business Recommendations

The analysis suggests three key recommendations. First, increase inventory before the holiday season since historical data consistently shows higher demand during November and December. Second, adopt the XGBoost forecasting model for monthly inventory planning because it produced the most accurate predictions among all evaluated models. Finally, implement demand-based stocking policies instead of maintaining the same inventory strategy for every product category, allowing resources to be allocated more efficiently.

Limitation

The forecasting system relies primarily on historical sales patterns. Unexpected events such as economic changes, supply chain disruptions, competitor actions, or large promotional campaigns cannot be predicted accurately. Therefore, the forecasting model should be updated regularly with new sales data to maintain its accuracy and business value.

Conclusion

This project demonstrates that combining exploratory data analysis, machine learning forecasting, anomaly detection, and product demand segmentation provides meaningful business insights for retail inventory management. The proposed solution enables more accurate demand forecasting, supports better stocking decisions, and helps improve operational efficiency. By adopting these recommendations, the business can reduce inventory costs, improve product availability, and make more informed strategic decisions.